

A Study on Neural Style Transfer Methods for Images

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Abstract— Image style transfer techniques have been around for almost twenty years. In particular, the rise of deep learning has gradually influenced traditional image transfer techniques, eventually giving rise to new neural image transfer techniques. This paper covers an introduction to traditional style transfer techniques, including Stroke-based rendering, image filtering, image analogy and texture synthesis, as well as a categorization of various current classic neural image style transfer methods including slow neural method, fast neural method and generative adversarial networks (GAN) based neural style transfer method. In addition, this paper covers commonly used training datasets for style transfer. Finally, the challenges of neural image style transfer are discussed, and the possible future research directions are contained.

Keywords—neural style transfer; deep learning; convolutional neural network; image rendering; generative adversarial nets; texture synthesis

I. INTRODUCTION

Style transfer normally indicates the transfer of the characteristic style of one art carrier to other works, which covers art carriers including text, sound, image, etc. As shown in Fig 1, image style transfer is to extract the features in the style image and the content of the target image to synthesize a new image with the features of the style image and the target image. There are countless artistic styles in the field of images. For a long period, people have spent a lot of time studying a classic painting, trying to reproduce the mastery of the great ancient painting style. As computer performance surged in the 20th century, people turned to the field of deep learning, and image style transfer was born. Neural style transfer networks gradually replaced manual image rendering. Now, through neural image style transfer techniques, we can easily obtain Van Gogh-like art style images to be used as a reference for learning and making more creative image styles. Early style transfer techniques mainly include non-photorealistic rendering (NPR) and texture compositing. NPR can be further divided into Stroke-based rendering, Image analogy and Image filtering according to its rendering characteristics.

A. Stroke-based rendering (SBR)

Meier [1] proposed a stroke-based method for rendering animations. A reference image is used to define the 2d stroke properties with particles to define the position of the stroke rendering for the animation. Hertzmann [2] proposed a new image generation algorithm. Create long, curved strokes by moving in the direction of the image gradient and select the brush stroke size to render in an adjustable rendering style. The SBR method is usually optimized for the simulation of a single

style. Although it is better to simulate specific types of styles such as oil painting, watercolor pencils, etc., the SBR method cannot simulate any style without a special design.

B. Image filtering

Image filtering was originally used for image noise reduction and due to the demand for light style transfer in industries such as film and television, image filtering was proposed as a light style transfer method. Tomasi [3] et al. first proposed bilateral filtering, which is a significant improvement on Gaussian filtering. The weighting coefficients of bilateral filtering in neighborhood filtering take into account not only geometric distance but also grayscale similarity. Winnemöller [4] et al. proposed the use of Gaussian filter differences to generate animation styles. Various filters have been derived since then. This style transfer method is fast and reliable, but since it is usually filtered globally, it does not simulate local features well and has disadvantages in terms of style diversity.

C. Image analogy

In 2001, Hertzmann [5] et al. first proposed the image analogy framework by using a pair of images as training samples in the design phase, where one image is a filtered version of the other image. The image filter parameters are obtained by learning pixel-by-pixel and applied to a new target image to produce a new image of a similar style. The idea of this is very close to the current neural style transfer. Later, Liao [6] et al. innovatively proposed to obtain better imaging results by using deep convolutional neural networks (dCNN) to extract features in the deep feature space and to perform matching. In fact, image analogy obtained good generative results after combining deep learning methods, but it is difficult to be widely used in practice due to the demanding image pairing requirements.

D. Texture synthesis

We usually think of artwork rendering as a branch of texture processing. Texture synthesis focuses more on texture analysis using statistical methods. One of the most successful ones is Efros [7] et al. proposed MRFs-based texture synthesis, where each pixel is filled by estimating the conditional distribution of pixels one by one through a Markov random field model by neighborhood search. Kwatra [8] et al. proposed an MRF optimization algorithm that defines a texture energy function and attains higher image quality by minimizing the energy function. Ashikhmin [9] et al. proposed a model based on MRF fast texture conversion to solve the drawback of too slow pixel-by-pixel filling.

Traditional style transfer techniques have been able to address some of the needs of industry and made some exploration of image features before neural style transfer techniques. However, the traditional style transfer technique commonly extracts the image features at the lower level and does not experiment with the higher-level features of the image. Moreover, traditional style transfer is normally performed by analyzing pixel-by-pixel, which is inefficient to compile. The convolutional neural network (CNN) gradually becoming a new direction in the field of style transfer with its powerful object recognition and deep semantic mining ability. In 2015, Gatys [10] et al. novelty proposed a neural network-based approach to style transfer. This method generates stylized original content images by iteratively training a pre-trained VGG (Visual Geometry Group) [11] network to get the high-level features of images, which are then matched to random noise maps. Since then, the door to neural style transfer has been opened.

In this paper, we systematically review the origin and development process of image neural style transfer. We classify neural style transfer according to its developmental history and evaluate each classification. Since there are many approaches to neural style transfer, this paper also provides the advantages and disadvantages of the mainstream approaches. Furthermore, several common training datasets are presented. Finally, this paper summarizes the current problems of image neural style transfer and provides an outlook on future works.

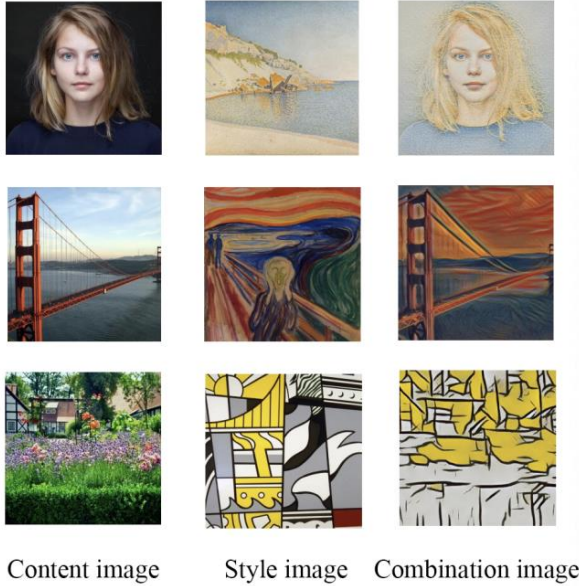


Fig. 1 Example of Style Transfer [12]

II. NEURAL STYLE TRANSFER METHODS FOR IMAGES

A. Slow Online Image Optimization Neural Method

This type of algorithm is very relevant to texture modelling methods. One branch is the summary statistics-based parametric slow neural method with Summary Statistics, represented by the Gram matrix proposed by Gatys [13] et al. Another algorithm is

Markov Random Fields (MRFs) based non-parametric slow neural method.

1) Summary Statistics-based Parametric Slow Neural Method

This kind of method first performs texture modelling using a parametric method based on summary statistics and then iterates gradient descent on the image pixel space to obtain the minimization objective function. This method is less efficient because it performs multiple iterations of optimization for each reconstructed image. The method originated from a CNN based texture synthesis method proposed by Gatys [14] et al. in 2015.

Gatys [14] et al. proposed two loss functions: style image loss and content image loss, which is the first time turned the image style transfer problem into a problem of finding the total loss function minimization. The total image loss function is given by:

$$\mathcal{L}_{total}(p, a, x) = \alpha \mathcal{L}_{content}(p, x) + \beta \mathcal{L}_{style}(a, x) \quad (1)$$

Where p represents the content image, a represents the style image, and the Synthesized image is represented by x ; β and α correspond to the weights of the style image losses and the content image losses of the total loss function. The following definition is the content loss function:

$$\mathcal{L}_{content}(p, x, l) = \frac{1}{2} \sum_{i,j} (F_{ij}^l - P_{ij}^l)^2 \quad (2)$$

l denotes layer l of the VGG network. F_{ij}^l and P_{ij}^l respectively represent the j activation value of the style image and content image at i feature map of the VGG network. In fact, the content loss function is represented by the squared error. Next is the definition of the style loss function:

$$\mathcal{L}_{style}(a, x) = \sum_{l=0}^L w_l E_l \quad (3)$$

$$E_l = \frac{1}{4N_l^2 M_l^2} \sum_{i,j} (G_{ij}^l - A_{ij}^l)^2 \quad (4)$$

N_l represents the l layer with N_l feature mapping where each size is M_l . G_{ij}^l and A_{ij}^l represent the style representation of the style image and content image in the l layer. The style representation is defined by the Gram matrix:

$$G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l \quad (5)$$

Gatys [14] et al. proposed to construct a unique feature space using a Gram matrix. This feature space is structurally free and can be constructed on top of any layer of filter responses. Gram matrix can successfully extract image styles and represents the stylistic information of an image. But still cannot interpret properties such as brush strokes, image semantic information, etc. Based on the Gatys [14] et al. works Berger and Memisevic [15] achieved global symmetry in texture generation by adding Markov structures to establish feature constraints. Gatys [14] et al. only considered the case of second-order moments. Kalischek [16] et al. proposed an optimization framework based on central moment discrepancy (CMD) by extending first-order moments

and second-order moments to higher orders before moment matching to better separate images from a semantic perspective.

2) MRFs Based Non-parametric Slow Neural Method

The non-parametric technique performs texture synthesis by extracting pixels from a resampled reference image [17]. Excellent methods for non-parametric techniques include deep image analogy and Markov random fields. Early Markov random field texture synthesis to extract image features is generally obtained by the nearest-neighbor scheme, by learning the features of that patch at local pixel points [17]. However, this leads to distortion of the image of that patch. Li et al [18]. proposed a novel style transfer method combining a deep convolutional neural network (dCNN) and a Markov random field. This method first argues that the powerful feature extraction capability of CNN should be used instead of the nearest neighbor algorithm to extract features. However, the Gram matrix was abandoned due to the problem of artifacts caused by the same computation result for different textures. Instead, MRFs are used to limit the generated image space and the problem of the Gram matrix is also solved. Because MRFs use style patch matching which is closer to the original image. Nevertheless, the MRFs method still has the caustic requirement that the datasets are similar in shape. Moreover, the MRFs method is not fast because it has to perform patch matching.

B. Fast Offline Model Optimization Neural Method.

Fast neural style transfer is based on feed-forward style networks to generate images, which are stylized by offline model generators. Based on the image generation model it can be further split into Per-Style-Per-Model, Multiple-Style-Per-Model and Arbitrary-Style-Per-Model Fast Neural Methods.

Per-Style-Per-Model is to design a specific style model mainly through the feed-forward network, and only need to input content images can quickly generate style images. In the previous work by Gatys [14] et al. each image requires throughout a separate pre-training optimization network which causes more time cost. Johnson [19] et al. propose a perceptual optimization model. The article uses a feed-forward CNN based on the perceptual loss to accomplish the style transfer and super-resolution work. Training the feedforward network with a perceptual loss function was successfully achieved much faster than Gatys [14] et al. In the same period, a similar feed-forward stylized transfer network was proposed by Ulyanov et al. The paper follows part of the model of Gatys [14] et al. such as the Gram matrix. They innovatively used a multi-scale architecture similar to the GAN network as a generative network so that they could process images of arbitrary size. Later, Ulyanov [20] et al. found in their experiments that the effect of image contrast was mainly due to the style images rather than the content images. They proposed an Instance normalization (IN) [21] method, which has a better convergence effect than the widely used Batch normalization and it can remove the effect of content images on image contrast.

Multiple-Style-Per-Model primarily refers to patterns that use a single model to solve multiple styles as output. Dumoulin [12] et al. consider that although feedforward-based stylized networks solve the problem of network generation speed, the single application scenario, and the fact that n styles correspond

to n training network parameters are unaffordable for devices with small memory. They believe that there is also commonality between different art styles. For example, there are similarities in the brush strokes of many artworks. So, they proposed conditional instance normalization (CIN). The CIN adds an affine transformation to the IN. The function of storing different images is achieved by transforming a small group of CIN parameters. A similar approach to style separation was proposed by Chen [22] et al. separating the content from the style network. On the one hand, the content network restores the original image as much as possible by means of an auto-encoder. On the other hand, the style network consists of an auto-encoder and a style bank layer, where each bank filter stores a style. The network can therefore separate the content image from the style image as much as possible and generate multiple style images. Li [23] et al. proposed a model for multi-style generation using N -dimensional one-hot encoding as the style selection unit.

Arbitrary-Style-Per-Model was inspired by previous work on whether it is possible to generate arbitrary models without training. The idea was first proposed by Chen [24] et al. They constructed a style-swap layer and used a pre-trained VGG network to match the extracted content feature blocks with the style feature blocks by computing nearest-neighbor matching. The result of the style-swap layer is then fed into the transposed convolution to decode the style image. This method has faster image iteration convergence than the method of Gatys [14] et al. However, as the processing of the style-swap layer still takes some time, the method is not yet able to achieve real-time image style transfer. A new arbitrary style transfer method was proposed by Huang [25] et al. called Adaptive Instance Normalization (AdaIN) which is the first time achieved real-time style transfer. Their work also received inspiration from IN but did not set affine parameters that could be learned as in CIN. AdaIN used mapping parameters that were computed adaptively. Combining the distribution of content image features with the distribution of style image features in which demonstrate that the feature transformation is finished in feature space. Thus, it could achieve real-time results without a training process. Park [26] et al. found that previous arbitrary style transfers existed with an imbalance between global and local styles. They proposed a self-attention mechanism to keep the global and local styles uniform based on the semantic spatial distribution by increasing the identity loss and multi-level feature embeddings. Li [27] et al. proposed a feature transformation-based image style transfer technique similar to AdaIN. The authors concluded that the extraction of image features can be easily achieved by performing whitening and coloring transformations (WCT) on the classical signals on the features. And the whitening and coloring change operations can be achieved by aligning the covariance matrix of the content image with the covariance matrix of the style image. The WCT method suffers from image artefacts. Yoo [28] et al. proposed the WCT2 model which preserves more structural information and improves image reality by turning the pooling layer into a wavelet pooling based on the WCT model.

C. A Generative adversarial networks-based method for image neural style transfer

Generative adversarial networks (GAN) [29] were proposed as a new approach to image generation. GAN comprise a generator and a discriminator. GAN is based on the following principles:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim P_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log (1 - D(G(z)))] \quad (6)$$

G and D correspond to the generator and discriminator, x means the real data, P_{data} can be seen as the data probability density distribution, z represents random Gaussian noise and $P_z(z)$ represent the prior probability of noise variable. The generator takes the introduced random gaussian noise and generates fake images to attempt to cheat the discriminator. Afterwards, the discriminator determines whether the image comes from the generator or not. Unlike the static gradient descent Optimisation of previous CNN structure, GAN optimize each other in dynamic adversarial situations.

Li [30] et al. had an idea that they could improve the speed of previous deep MRFs texture synthesis [17] more than 500 times by using a discriminative network in GAN instead of patch matching. Mirza [31][31] et al. propose that CGAN hopes to control the GAN to generate specific images rather than random images through conditional probability. The core of this is the addition of the conditional attribute constraint y to the GAN, where y can be any label information. CGAN successfully brings GAN into the domain of supervised learning. Zhu [32] et al. were inspired by CGAN and proposed a pix-2-pix model. First, they prepared strictly paired source data x and target data y . They input the target image y and the source image x into the discriminator D . The discriminator D is trained with the target image y and the source image x . Afterwards, the generated images are passed through the discriminator to determine whether they are true or false. They creatively use U-Net [33] as the model structure instead of the traditional Multi-layer Perceptron (MLP) structure. In the field of image translation, pix-2-pix has achieved great results. However, the data set for this method is too demanding to be used in practice due to the need for strictly paired images. Zhu [34] et al. proposed a new arbitrary style transfer model called CycleGAN to address the previous shortcomings. The CycleGAN uses two discriminators D_X and D_Y and two generators G and F . For a given distribution X and Y , D_X is used to determine the content image and D_Y is used to determine the style image. The generators G and F are mappings of X to Y and Y to X respectively. The generated picture u needs to cheat the discriminators D_X and D_Y as much as possible at the same time, and eventually reach the global optimum in iterative learning. The authors propose a loss function called Cycle Consistency Loss to let the output image of the generator differs from the input image only in style but has the same content. Choi [35] et al. proposed StarGAN, which can switch between multiple domains by learning only one model.

In the field of image style transfer, GAN has led to a number of models, which significantly improve the speed of image generation and also provide good image quality. GAN is already widely used in industry and has promising commercial value.

D. Datasets

1) Microsoft COCO Dataset

MS COCO is the largest dataset with semantic segmentation to date. Over 330,000 images are provided, 200,000 of which are labelled. Paper [19][22][24][25][26][28] are trained based on this dataset.

2) ImageNet Dataset

The ImageNet[36] dataset is a classic and widely used dataset for image classification. ImageNet has the largest image data size of over 14 million images in over 20,000 categories. Paper [12] is trained based on this dataset.

3) MIR Flickr 25K Dataset

The Flickr dataset has 25,000 images. The dataset is distinct in that each image has a corresponding tag and annotation. Paper [31] is trained based on this dataset.

4) Cityscapes

The Cityscape dataset has a larger number of images for road driving than the COCO dataset. The dataset consists of 19 categories, 5000 fully annotated images and 20,000 non-fully annotated images. Paper [32] are trained based on this dataset.

5) Describable Textures Dataset (DTD)

The DTD dataset is a texture-focused database containing 5640 images. DTD has 47 categories and over 120 images per category. Paper [23] is trained based on this dataset.

III. CONCLUSION AND FUTURE PROSPECTS

Neural-based image style transfer techniques have become relatively mature. We can find image transfer gradually moving from theory to commercial implementation. However, there are still some challenges with neural image style transfer. First of all, a standardized image evaluation criterion for neural image style transfer is still lacking. Different style transfer methods usually correspond to customized image evaluation criteria. Most current methods can only evaluate image quality through human assessment, which is highly subjective. The second challenge is that the convolutional neural networks commonly used for feature extraction during pre-training take too long to train and are prone to mode collapse.

In general, image neural style transfer has been applied to many scenarios and has great potential for development. Further research to address the current imperfect evaluation methods and the semantic ambiguity of the generated images may be able to improve image quality. At the same time, most image transfer methods normally introduce random Gaussian noise, which makes it difficult to let the generated images controllable. The targeted personalization of images is one of the future research directions.

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